Q1 Language of kinematics

Name:

Class:

Date:

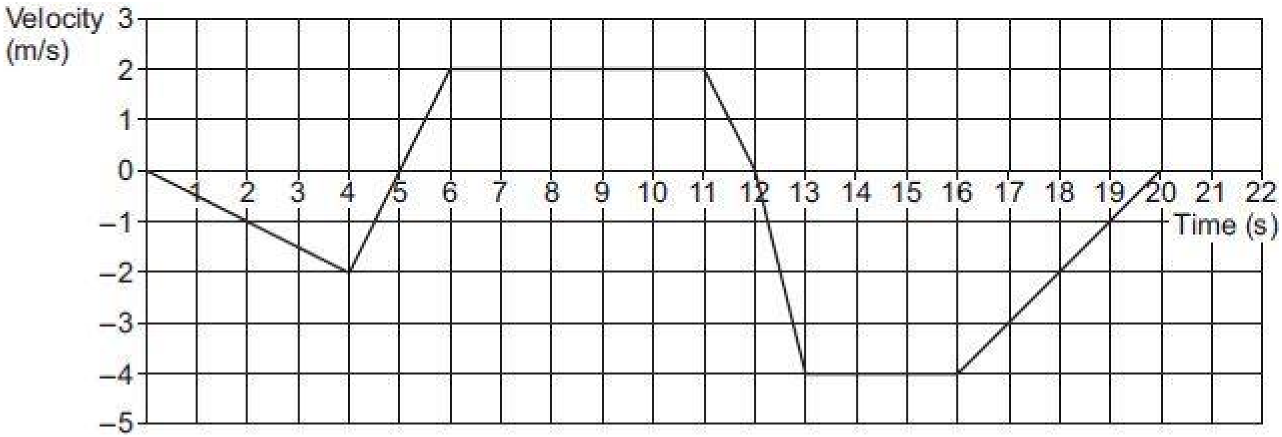
Time: **433 minutes**

Marks: **360 marks**

Comments:

Q1.

The graph below shows the velocity of an object moving in a straight line over a 20 second journey.



(a) Find the maximum magnitude of the acceleration of the object.

(1)

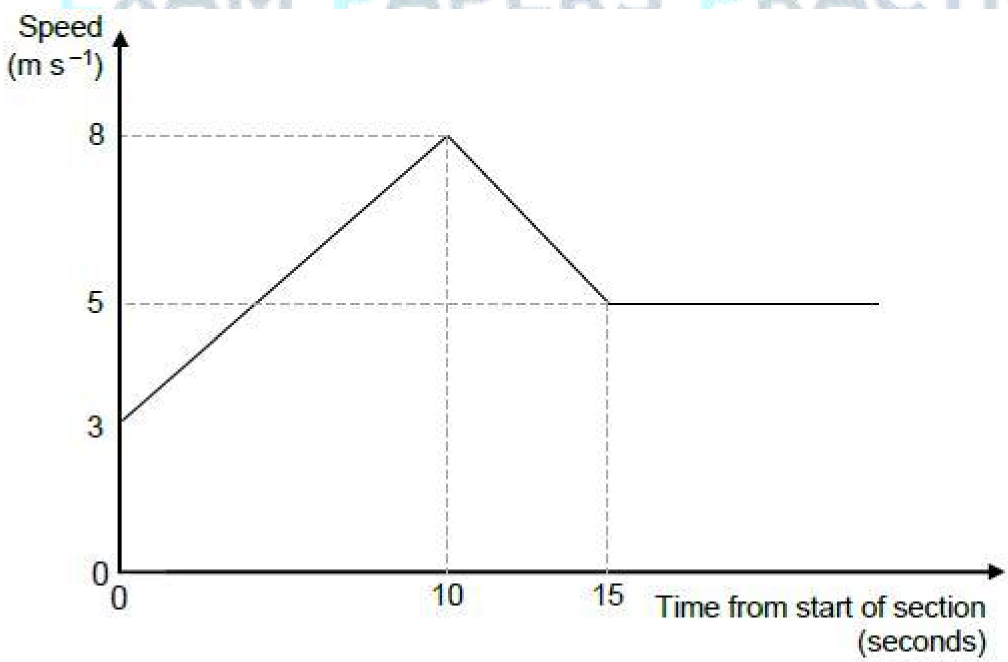
(b) The object is at its starting position at times 0, t_1 and t_2 seconds.

Find t_1 and t_2

(4)
(Total 5 marks)

Q2.

The graph shows how the speed of a cyclist varies during a timed section of length 120 metres along a straight track.



(a) Find the acceleration of the cyclist during the first 10 seconds.

(1)

(b) After the first 15 seconds, the cyclist travels at a constant speed of 5 m s^{-1} for a further T seconds to complete the 120-metre section.

Calculate the value of T.

(4)

(Total 5 marks)

Q3.

A particle, of mass 400 grams, is initially at rest at the point O.

The particle starts to move in a straight line so that its velocity, v m s^{-1}, at time t seconds is given by

$$v = 6t^2 - 12t^3 \text{ for } t > 0$$

(a) Find an expression, in terms of t, for the force acting on the particle.

(3)

(b) Find the time when the particle next passes through O.

(5)

(Total 8 marks)

Q4.

A single force of magnitude 4 newtons acts on a particle of mass 50 grams.

Find the magnitude of the acceleration of the particle.

Circle your answer.

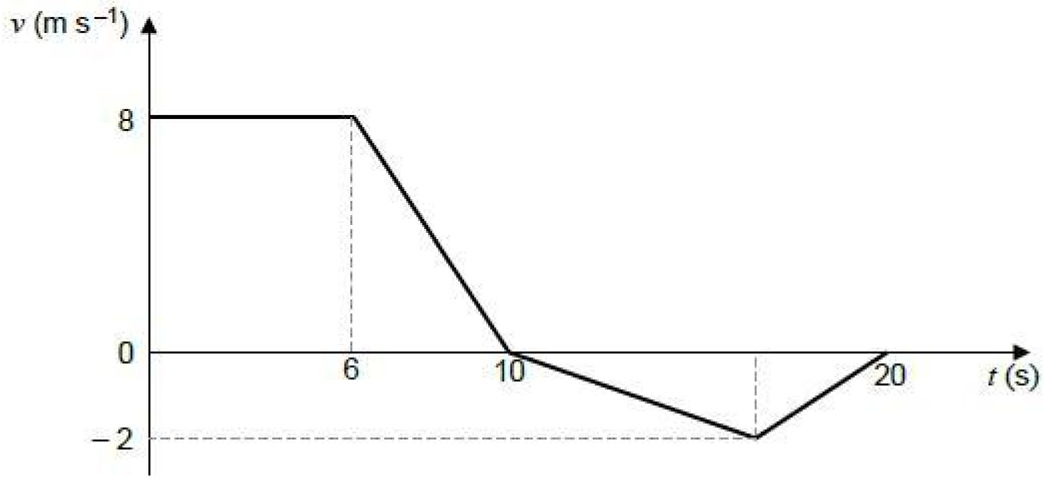
12.5 m s^{-2} 0.08 m s^{-2} 0.0125 m s^{-2} 80 m s^{-2}

(Total 1 mark)

Q5.

The graph below models the velocity of a small train as it moves on a straight track for 20 seconds.

The front of the train is at the point A when $t = 0$
The mass of the train is 800kg.



(a) Find the total distance travelled in the 20 seconds.

(3)

(b) Find the distance of the front of the train from the point *A* at the end of the 20 seconds.

(1)

(c) Find the maximum magnitude of the resultant force acting on the train.

(2)

(d) Explain why, in reality, the graph may not be an accurate model of the motion of the train.

(1)

(Total 7 marks)

Q6.

At time $t = 0$, a parachutist jumps out of an airplane that is travelling horizontally.
The velocity, $\mathbf{v}$ m s^{-1}, of the parachutist at time t seconds is given by:

$$\mathbf{v} = (40e^{-0.2t})\mathbf{i} + 50(e^{-0.2t} - 1)\mathbf{j}$$

The unit vectors **i** and **j** are horizontal and vertical respectively.
Assume that the parachutist is at the origin when $t = 0$
Model the parachutist as a particle.

(a) Find an expression for the position vector of the parachutist at time t.

(4)

(b) The parachutist opens her parachute when she has travelled 100 metres horizontally.

Find the vertical displacement of the parachutist from the origin when she opens her parachute.

(4)

(c) Carefully, explaining the steps that you take, deduce the value of g used in the formulation of this model.

(3)
(Total 11 marks)

Q7.

In this question use g = 9.81 m s^{-2}, giving your final answers to an appropriate degree of accuracy.

A ball is projected from the origin. After 2.5 seconds, the ball lands at the point with position vector (40**i** – 10**j**) metres.
The unit vectors **i** and **j** are horizontal and vertical respectively.
Assume that there are no resistance forces acting on the ball.

(a) Find the speed of the ball when it is at a height of 3 metres above its initial position.

(6)

(b) State the speed of the ball when it is at its maximum height.

(1)

(c) Explain why the answer you found in part **(b)** may not be the actual speed of the ball when it is at its maximum height.

(1)
(Total 8 marks)

Q8.

A car is travelling at a speed of 20 m s^{-1} along a straight horizontal road. The driver applies the brakes and a constant braking force acts on the car until it comes to rest.

(a) Assume that no other horizontal forces act on the car.

(i) After the car has travelled 75 metres, its speed has reduced to 10 m s^{-1}. Find the acceleration of the car.

(3)

(ii) Find the time taken for the speed of the car to reduce from 20 m s^{-1} to zero.

(2)

(iii) Given that the mass of the car is 1400 kg, find the magnitude of the constant braking force.

(2)

(b) Given that a constant air resistance force of magnitude 200 N acts on the car during the motion, find the magnitude of the constant braking force.

(1)
(Total 8 marks)

Q9.

A particle moves with a constant acceleration of (0.1**i** − 0.2**j**) m s^{-2}. It is initially at the origin where it has velocity (−**i** + 3**j**) m s^{-1}. The unit vectors **i** and **j** are directed east and north respectively.

(a) Find an expression for the position vector of the particle t seconds after it has left the origin.

(2)

(b) Find the time that it takes for the particle to reach the point where it is due east of the origin.

(3)

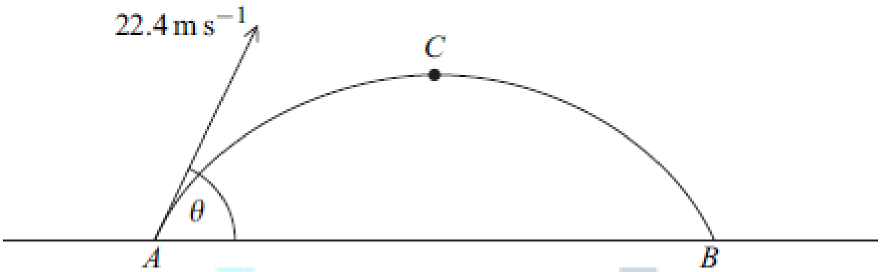
(c) Find the speed of the particle when it is travelling south-east.

(6)

(Total 11 marks)

Q10.

A particle is launched from the point A on a horizontal surface, with a velocity of 22.4 m s^{-1} at an angle θ above the horizontal, as shown in the diagram.



After 2 seconds, the particle reaches the point C, where it is at its maximum height above the surface.

(a) Show that $\sin \theta = 0.875$.

(3)

(b) Find the height of the point C above the horizontal surface.

(3)

(c) The particle returns to the surface at the point B. Find the distance between A and B.

(3)

(d) Find the length of time during which the height of the particle above the surface is greater than 5 metres.

(5)

(e) Find the minimum speed of the particle.

(2)

(Total 16 marks)

Q11.

A particle moves in a straight line. At time t seconds, it has velocity v m s^{-1}, where

$$v = 6t^2 - 2e^{-4t} + 8$$

and $t \geq 0$.

(a) (i) Find an expression for the acceleration of the particle at time t.

(2)

(ii) Find the acceleration of the particle when $t = 0.5$.

(2)

(b) The particle has mass 4 kg.

Find the magnitude of the force acting on the particle when $t = 0.5$.

(1)

(c) When $t = 0$, the particle is at the origin.

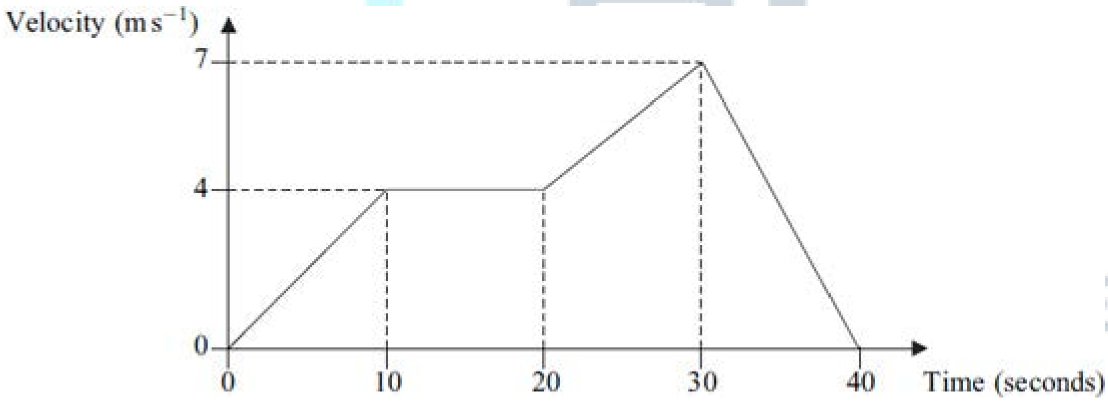
Find an expression for the displacement of the particle from the origin at time t.

(4)

(Total 9 marks)

Q12.

The graph shows how the velocity of a train varies as it moves along a straight railway line.



(a) Find the total distance travelled by the train.

(4)

(b) Find the average speed of the train.

(2)

(c) Find the acceleration of the train during the first 10 seconds of its motion.

(2)

(d) The mass of the train is 200 tonnes. Find the magnitude of the resultant force acting on the train during the first 10 seconds of its motion.

(2)

(Total 10 marks)

Q13.

A particle moves with constant acceleration $(-0.4\mathbf{i} + 0.2\mathbf{j})$ m s^{-2} . Initially, it has velocity $(4\mathbf{i} + 0.5\mathbf{j})$ m s^{-1} . The unit vectors **i** and **j** are directed east and north respectively.

(a) Find an expression for the velocity of the particle at time t seconds.

(2)

(b) (i) Find the velocity of the particle when $t = 22.5$.

(2)

(ii) State the direction in which the particle is travelling at this time.

(1)

(c) Find the time when the speed of the particle is 5 m s^{-1}.

(6)

(Total 11 marks)

Q14.

A crane is used to lift a load, using a single vertical cable which is attached to the load. The load accelerates uniformly from rest. When it has risen 0.9 metres, its speed is 0.6 m s^{-1}.

(a) (i) Show that the acceleration of the load is 0.2 m s^{-2}.

(3)

(ii) Find the time taken for the load to rise 0.9 metres.

(2)

(b) Given that the mass of the load is 800 kg, find the tension in the cable while the load is accelerating.

(3)

(Total 8 marks)

Q15.

A pair of cameras records the time that it takes a car on a motorway to travel a distance of 2000 metres. A car passes the first camera whilst travelling at 32 m s^{-1}.
The car continues at this speed for 12.5 seconds and then decelerates uniformly until it passes the second camera when its speed has decreased to 18 m s^{-1}.

(a) Calculate the distance travelled by the car in the first 12.5 seconds.

(1)

(b) Find the time for which the car is decelerating.

(3)

(c) Sketch a speed–time graph for the car on this 2000-metre stretch of motorway.

(3)

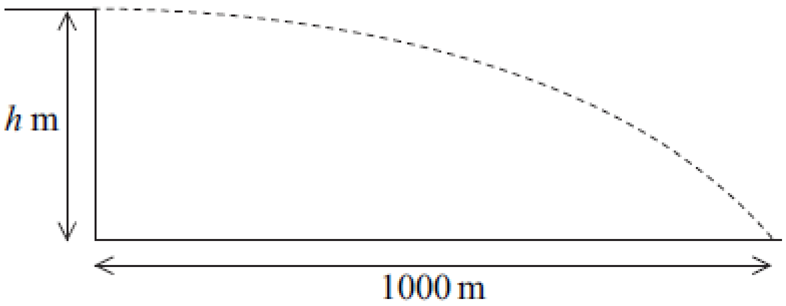
(d) Find the average speed of the car on this 2000-metre stretch of motorway.

(2)

(Total 9 marks)

Q16.

A bullet is fired horizontally from the top of a vertical cliff, at a height of h metres above the sea. It hits the sea 4 seconds after being fired, at a distance of 1000 metres from the base of the cliff, as shown in the diagram.



(a) Find the initial speed of the bullet.

(2)

(b) Find h.

(2)

(c) Find the speed of the bullet when it hits the sea.

(4)

(d) Find the angle between the velocity of the bullet and the horizontal when it hits the sea.

(3)

(Total 11 marks)

Q17.

A helicopter is initially hovering above a lighthouse. It then sets off so that its acceleration is $(0.5\mathbf{i} + 0.375\mathbf{j})$ m s^{-2}. The helicopter does not change its height above sea level as it moves. The unit vectors **i** and **j** are directed east and north respectively.

(a) Find the speed of the helicopter 20 seconds after it leaves its position above the lighthouse.

(4)

(b) Find the bearing on which the helicopter is travelling, giving your answer to the nearest degree.

(3)

(c) The helicopter stops accelerating when it is 500 metres from its initial position.

Find the time that it takes for the helicopter to travel from its initial position to the point where it stops accelerating.

(5)

(Total 12 marks)

Q18.

A particle moves in a horizontal plane under the action of a single force, **F** newtons.

The unit vectors **i** and **j** are directed east and north respectively. At time t seconds, the velocity of the particle, **v** m s^{-1}, is given by

$$\mathbf{v} = 4e^{-2t}\,\mathbf{i} + (6t - 3t^2)\mathbf{j}$$

(a) Find an expression for the acceleration of the particle at time t.

(3)

(b) The mass of the particle is 5 kg.

(i) Find an expression for the force **F** acting on the particle at time t.

(2)

(ii) Find the magnitude of **F** when $t = 0$.

(2)

(c) Find the value of t when **F** acts due west.

(2)

(d) When $t = 0$, the particle is at the point with position vector (6**i** + 5**j**) m.

Find the position vector, **r** metres, of the particle at time t.

(5)

(Total 14 marks)

Q19.

A ball is thrown vertically upwards at a speed of 4.9 m s^{-1} from a height of 1 metre above ground level. The ball is caught when it returns to a height of 1 metre above ground level and is travelling downwards.

(a) Find the time that it takes for the ball to reach its maximum height.

(3)

(b) Find the maximum height of the ball above ground level.

(3)

(c) State the total time for which the ball is moving.

(1)

(d) State the speed of the ball when it is caught.

(1)

(Total 8 marks)

Q20.

A boat moves with constant acceleration, so that its velocity **v** m s^{-1} at time t seconds is given by

$$\mathbf{v} = (4.5 - 0.9t)\mathbf{i} + (2 + 0.5t)\mathbf{j}$$

where the unit vectors **i** and **j** are directed east and north respectively.

(a) Find the velocity of the boat when $t = 0$.

(1)

(b) Find the velocity of the boat when $t = 5$.

(1)

(c) State the direction in which the boat is travelling when $t = 5$.

(1)

(d) Find the acceleration of the boat.

(3)

(e) The mass of the boat is 250 kg. Find the magnitude of the resultant force acting on the boat.

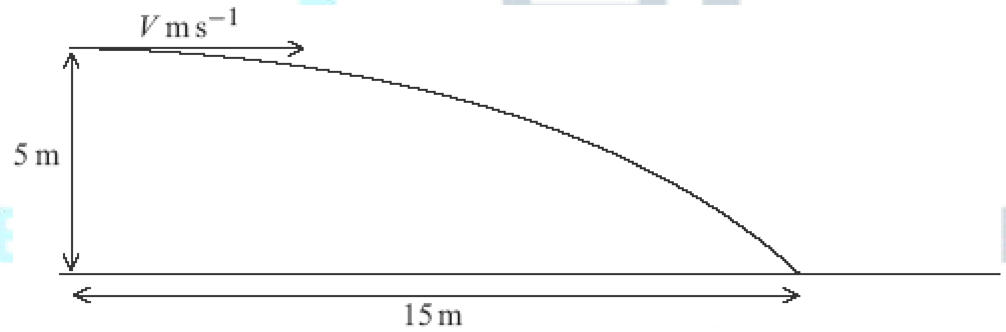
(3)

(f) Draw a diagram to show the direction of the resultant force on the boat and calculate the angle between this force and the unit vector **i**.

(4)
(Total 13 marks)

Q21.

A ball is projected horizontally with speed V m s^{-1} at a height of 5 metres above horizontal ground. When the ball has travelled a horizontal distance of 15 metres, it hits the ground.



(a) Show that the time it takes for the ball to travel to the point where it hits the ground is 1.01 seconds, correct to three significant figures.

(3)

(b) Find V.

(2)

(c) Find the speed of the ball when it hits the ground.

(4)

(d) Find the angle between the velocity of the ball and the horizontal when the ball hits the ground. Give your answer to the nearest degree.

(3)
(Total 12 marks)

Q22.

A sprinter accelerates from rest at a constant rate for the first 10 metres of a 100-metre race. He takes 2.5 seconds to run the first 10 metres.

(a) Find the acceleration of the sprinter during the first 2.5 seconds of the race.

(3)

(b) Show that the speed of the sprinter at the end of the first 2.5 seconds of the race is 8 m s^{-1}.

(2)

(c) The sprinter completes the 100-metre race, travelling the remaining 90 metres at a constant speed of 8 m s^{-1}. Find the total time taken for the sprinter to travel the 100 metres.

(3)

(d) Calculate the average speed of the sprinter during the 100-metre race.

(2)

(Total 10 marks)

Q23.

A ball is released from rest at a height of 15 metres above ground level.

(a) Find the speed of the ball when it hits the ground, assuming that no air resistance acts on the ball.

(3)

(b) In fact, air resistance does act on the ball. Assume that the air resistance force has a constant magnitude of 0.9 newtons. The ball has a mass of 0.5 kg.

(i) Draw a diagram to show the forces acting on the ball, including the magnitudes of the forces acting.

(1)

(ii) Show that the acceleration of the ball is 8 m s^{-2}.

(3)

(iii) Find the speed at which the ball hits the ground.

(2)

(iv) Explain why the assumption that the air resistance force is constant may not be valid.

(1)

(Total 10 marks)

Q24.

The constant forces $\mathbf{F}_1 = (8\mathbf{i} + 12\mathbf{j})$ newtons and $\mathbf{F}_2 = (4\mathbf{i} - 4\mathbf{j})$ newtons act on a particle. No other forces act on the particle.

(a) Find the resultant force acting on the particle.

(2)

(b) Given that the mass of the particle is 4 kg, show that the acceleration of the particle is (3**i** + 2**j**) m s^{-2}.

(2)

(c) At time t seconds, the velocity of the particle is **v** m s^{-1}.

(i) When t = 20, **v** = 40**i** + 32**j**.

Show that **v** = –20**i** – 8**j** when t = 0.

(3)

(ii) Write down an expression for **v** at time t.

(1)

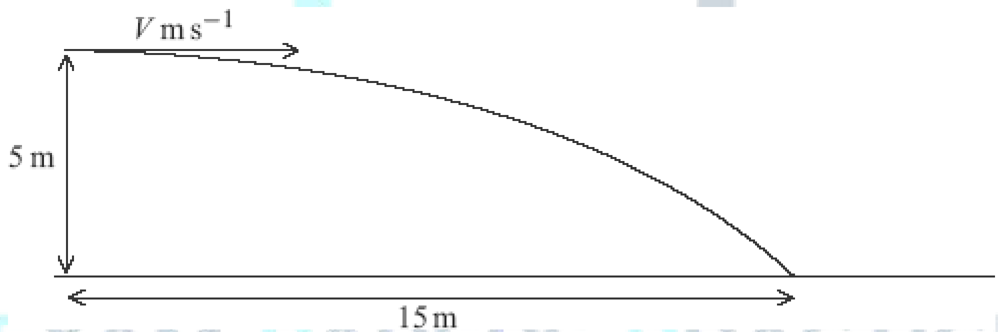
(iii) Find the times when the speed of the particle is 8 m s^{-1}.

(6)

(Total 14 marks)

Q25.

A ball is projected horizontally with speed V m s^{-1} at a height of 5 metres above horizontal ground. When the ball has travelled a horizontal distance of 15 metres, it hits the ground.



(a) Show that the time it takes for the ball to travel to the point where it hits the ground is 1.01 seconds, correct to three significant figures.

(3)

(b) Find V.

(2)

(c) Find the speed of the ball when it hits the ground.

(4)

(d) Find the angle between the velocity of the ball and the horizontal when the ball hits the ground. Give your answer to the nearest degree.

(3)

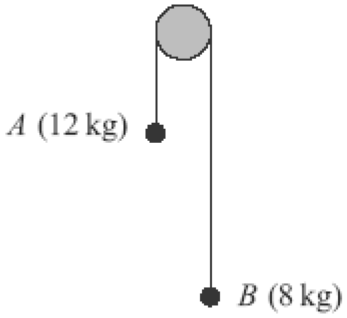
(e) State two assumptions that you have made about the ball while it is moving.

(2)

(Total 14 marks)

Q26.

Two particles, A and B, have masses 12 kg and 8 kg respectively. They are connected by a light inextensible string that passes over a smooth fixed peg, as shown in the diagram.



The particles are released from rest and move vertically. Assume that there is no air resistance.

(a) By forming two equations of motion, show that the magnitude of the acceleration of each particle is 1.96 m s^{-2}.

(5)

(b) Find the tension in the string.

(2)

(c) After the particles have been moving for 2 seconds, both particles are at a height of 4 metres above a horizontal surface. When the particles are in this position, the string breaks.

(i) Find the speed of particle A when the string breaks.

(2)

(ii) Find the speed of particle A when it hits the surface.

(3)

(Total 12 marks)

Q27.

A ship moves with constant acceleration. At time t seconds, its velocity is **v** m s^{-1}, where

$$\mathbf{v} = (9 - 0.01t)\mathbf{i} + (7 - 0.03t)\mathbf{j}$$

The unit vectors **i** and **j** are directed east and north respectively.

(a) Write down the velocity of the ship when $t = 0$.

(1)

(b) Find the acceleration of the ship.

(2)

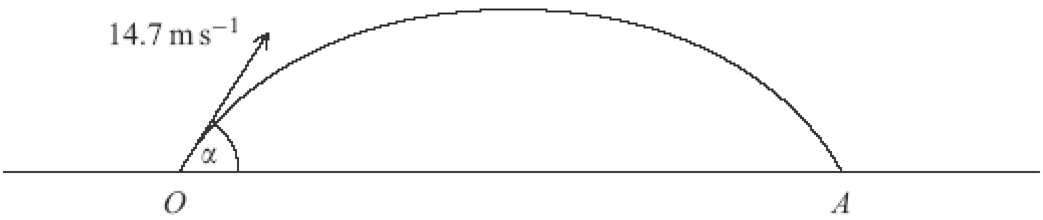
(c) Find t when the ship is travelling south-east.

(3)

(Total 6 marks)

Q28.

A ball is struck so that it leaves a horizontal surface travelling at 14.7 m s^{-1} at an angle α above the horizontal. The path of the ball is shown in the diagram.



(a) Show that the ball takes $\dfrac{3\sin\alpha}{2}$ seconds to reach its maximum height.

(3)

(b) The ball reaches a maximum height of 7 metres.

(i) Find α.

(5)

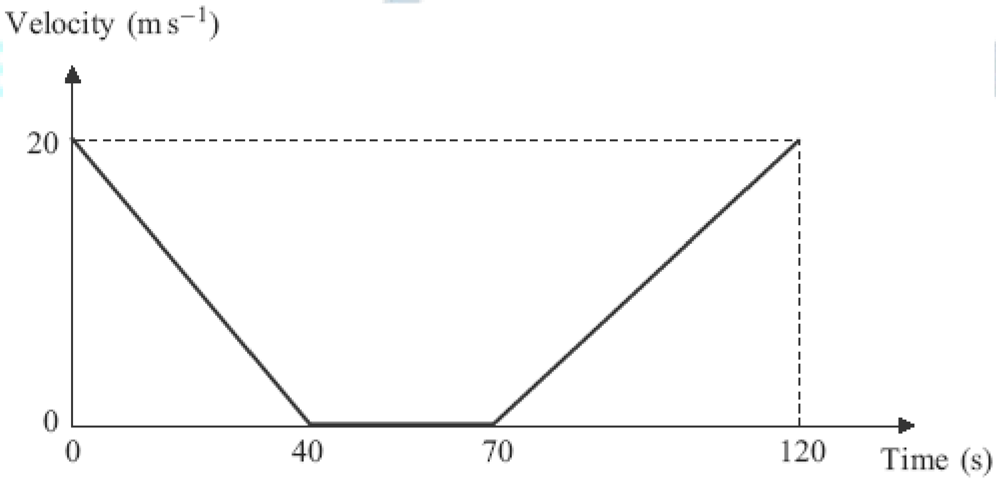
(ii) Find the range, OA.

(3)

(Total 11 marks)

Q29.

A bus slows down as it approaches a bus stop. It stops at the bus stop and remains at rest for a short time as the passengers get on. It then accelerates away from the bus stop. The graph shows how the velocity of the bus varies.



Assume that the bus travels in a straight line during the motion described by the graph.

(a) State the length of time for which the bus is at rest.

(1)

(b) Find the distance travelled by the bus in the first 40 seconds.

(2)

(c) Find the total distance travelled by the bus in the 120-second period.

(2)

(d) Find the average speed of the bus in the 120-second period.

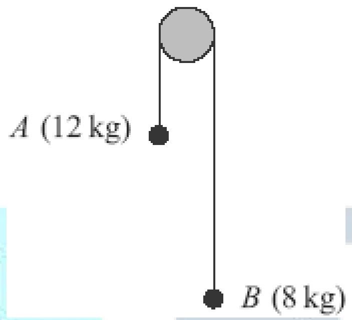
(2)

(e) If the bus had not stopped but had travelled at a constant 20 m s^{-1} for the 120-second period, how much further would it have travelled?

(2)
(Total 9 marks)

Q30.

Two particles, A and B, have masses 12 kg and 8 kg respectively. They are connected by a light inextensible string that passes over a smooth fixed peg, as shown in the diagram.



The particles are released from rest and move vertically. Assume that there is no air resistance.

(a) By forming two equations of motion, show that the magnitude of the acceleration of each particle is 1.96 m s^{-2}.

(5)

(b) Find the tension in the string.

(2)

(c) After the particles have been moving for 2 seconds, both particles are at a height of 4 metres above a horizontal surface. When the particles are in this position, the string breaks.

(i) Find the speed of particle A when the string breaks.

(2)

(ii) Find the speed of particle A when it hits the surface.

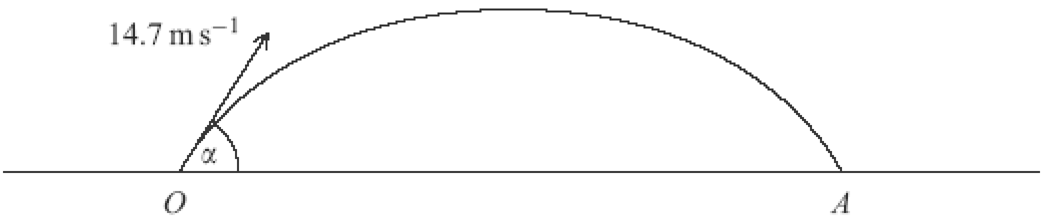
(3)

(iii) Find the time that it takes for particle B to reach the surface after the string breaks.
Assume that particle B does not hit the peg.

(5)
(Total 17 marks)

Q31.

A ball is struck so that it leaves a horizontal surface travelling at 14.7 m s^{-1} at an angle α above the horizontal. The path of the ball is shown in the diagram.



(a) Show that the ball takes $\dfrac{3\sin\alpha}{2}$ seconds to reach its maximum height.

(3)

(b) The ball reaches a maximum height of 7 metres.

(i) Find α.

(5)

(ii) Find the range, OA.

(3)

(c) State two assumptions that you needed to make in order to answer the earlier parts of this question.

(2)

(Total 13 marks)

Q32.

A particle has mass 200 kg and moves on a smooth horizontal plane. A single horizontal force, $\left(400\cos\left(\frac{\pi}{2}t\right)\mathbf{i} + 600t^2\,\mathbf{j}\right)$ newtons, acts on the particle at time t seconds.

The unit vectors **i** and **j** are directed east and north respectively.

(a) Find the acceleration of the particle at time t.

(2)

(b) When t = 4, the velocity of the particle is (–3**i** + 56**j**) m s^{-1}.

Find the velocity of the particle at time t.

(5)

(c) Find t when the particle is moving due west.

(3)

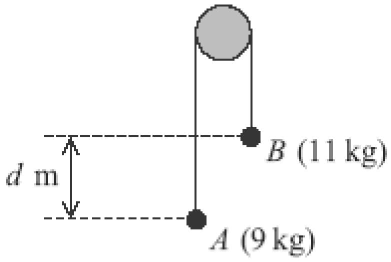
(d) Find the speed of the particle when it is moving due west.

(2)

(Total 12 marks)

Q33.

Two particles, A and B, are connected by a light inextensible string that passes over a smooth fixed peg, as shown in the diagram. The mass of A is 9 kg and the mass of B is 11 kg.



The particles are released from rest in the position shown, where B is d metres higher than A.
Assume that no resistance forces act on the particles.

(a) By forming an equation of motion for each of the particles A and B, show that the acceleration of each particle has magnitude 0.98 m s^{-2}.

(5)

(b) When the particles have been moving for 0.5 seconds, they are at the same level.

(i) Find the speed of the particles at this time.

(2)

(ii) Find d.

(4)

(Total 11 marks)

Q34.

A football is kicked from ground level with an initial velocity of 30 m s^{-1} at an angle of 35° above the horizontal.

(a) Find the maximum height of the ball above ground level.

(4)

(b) Show that when the speed of the ball is 28 m s^{-1}, the magnitude of the vertical component of its velocity is 13.4 m s^{-1}, correct to three significant figures.

(4)

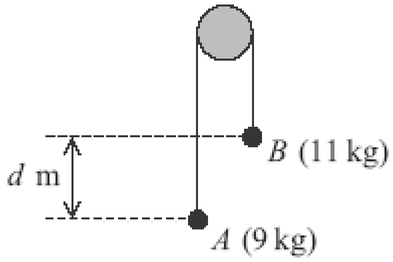
(c) Find the angle between the velocity of the ball and the horizontal when the speed of the ball is 28 m s^{-1}.

(2)

(Total 10 marks)

Q35.

Two particles, A and B, are connected by a string that passes over a fixed peg, as shown in the diagram. The mass of A is 9 kg and the mass of B is 11 kg.



The particles are released from rest in the position shown, where B is d metres higher than A.
The motion of the particles is to be modelled using simple assumptions.

(a) State one assumption that should be made about the peg.

(1)

(b) State two assumptions that should be made about the string.

(2)

(c) By forming an equation of motion for each of the particles A and B, show that the acceleration of each particle has magnitude 0.98 m s^{-2}.

(5)

(d) When the particles have been moving for 0.5 seconds, they are at the same level.

(i) Find the speed of the particles at this time.

(2)

(ii) Find d.

(4)

(Total 14 marks)